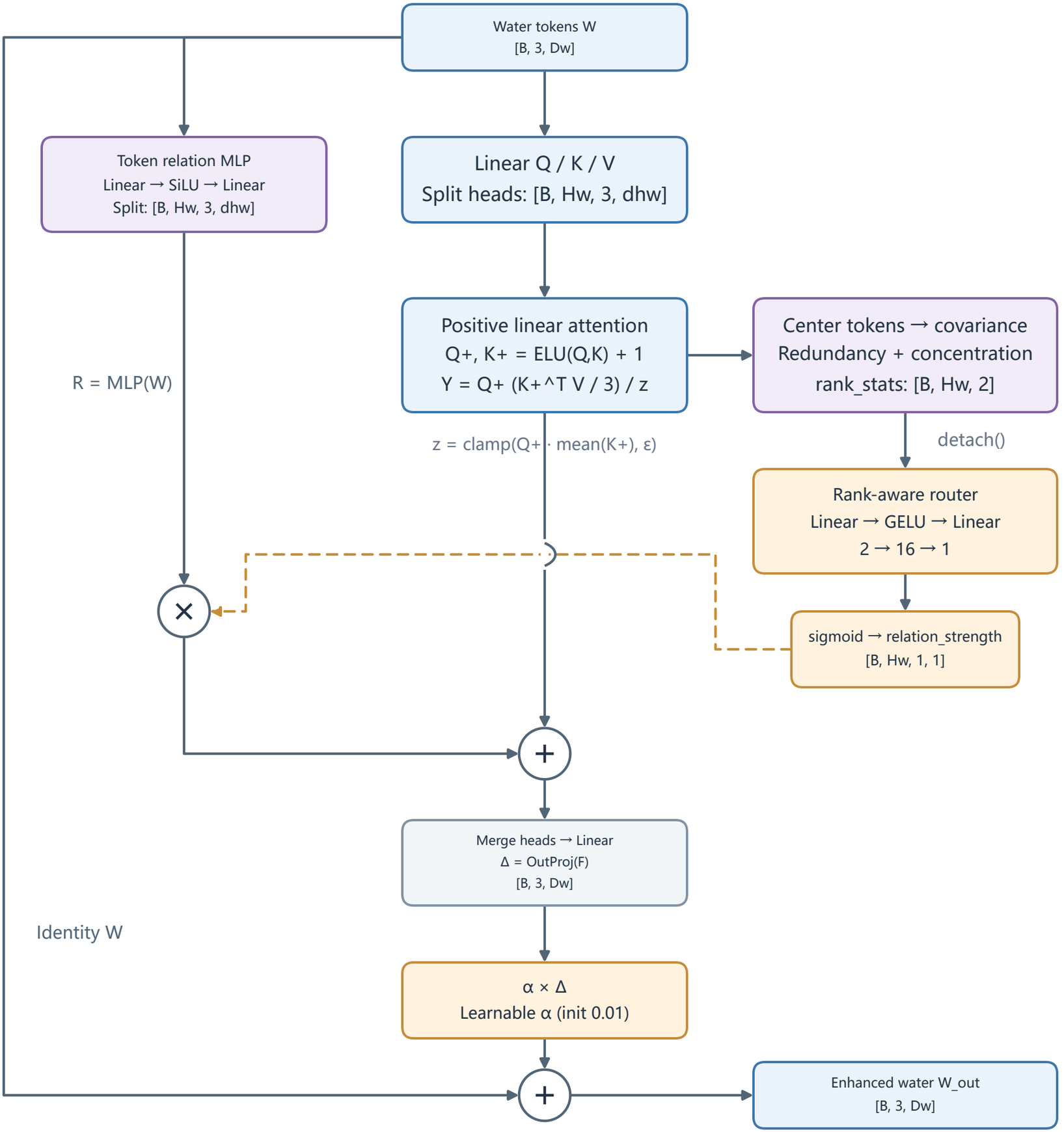
(a) Rank-aware Water Attention — RWA

水质关系建模 | Encoded water tokens in; enhanced water tokens out



Dw = 64, Hw = 4, dhw = 16; relation_strength controls the relation MLP; α scales the final residual.

Rank statistics are detached before the router. Spectral effective rank is diagnostic only.

Solid arrows: feature flow. Dashed arrows: learned controls. A wire jump is not a connection.